It's Not a Phase:
Predicting Frontier Alignment from Two Benchmark Scores

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Abstract

Frontier language model development is tracked through independent benchmark trajectories, obscuring whether capabilities reinforce or trade off across releases. We introduce an operational coupling diagnostic that maps paired benchmark scores into regime state and lab-relative residual on a 31-model, 8-lab frontier panel (2024–2026). On a curated core subset of 20 models, capability coupling is strongly cooperative ($r = +0.854$, $p < 10^{-5}$), and cross-lab holdout predicts GPQA scores within 6.9%. Each lab's training philosophy is readable as a signed residual diagnostic (h-field): Google is consistently reasoning-rich ($h = +5.7$), Anthropic coding-rich ($h = -6.7$), and DeepSeek shows a reversal trajectory (h: $+8.4 \to -4.4$) consistent with a shift in capability emphasis. We find preliminary evidence ($n = 4$) that SWE-bench performance is saturating while instruction-following activates as a new capability axis, consistent with a predicted capability-axis handoff at frontier scale. Seven falsifiable predictions with timestamped pass/fail criteria are provided. All scores are from public model cards and leaderboards; code and data are available at https://github.com/adilamin89/cape-scaling.

1 Introduction

When a frontier lab releases a new model, the community asks two questions: how well does it code, and how well does it reason? These questions are always asked separately. SWE-bench measures coding; GPQA Diamond measures reasoning; each gets its own leaderboard row, its own trajectory, its own narrative. But no one asks whether improving one helps or hurts the other—and at the frontier, this interaction turns out to be the more informative signal.

We test it. Using paired benchmark scores (SWE-bench Verified and GPQA Diamond) across 31 frontier models from 8 labs, we measure whether capabilities reinforce or undermine each other across a lab's release sequence. On a matched core set, capabilities cooperate ($r = +0.854$), but the pattern of cooperation varies by lab and over time. Each lab leaves a fingerprint—a one-number diagnostic that summarizes whether its models are becoming better reasoners, better coders, or both. Every frontier model carries a signature, and two public numbers are enough to read it.

This is a measurement-and-monitoring paper first: not a new benchmark, not a new model, but a new way to read frontier trajectory from public evaluations. We introduce CAPE (Capability Coupling Analysis of Phase Emergence), an operational diagnostic adapted for frontier measurement. It provides three outputs:

1. **Regime state**: cooperative vs. antagonistic coupling between capability axes.
2. **Lab-relative residual** (h-field): a signed diagnostic that summarizes how far each model deviates from the population cooperation trend.

3. **Transition risk**: preliminary signals that a new capability axis may be activating as an existing one saturates.

The base-model version of this framework establishes that capability coupling undergoes a regime transition near 3.5 billion parameters, confirmed across 63 models and 16 families [Amin(2026)]. Here we extend to the frontier, where all models sit deep in the cooperative regime but follow lab-specific trajectories through coupling space.

Unlike leaderboards that rank models on independent axes, CAPE measures the *relationship* between axes—revealing structure that no single-benchmark comparison can detect. A model that ranks first on SWE-bench and fifth on GPQA tells a different story than one that ranks third on both: CAPE captures this difference as a signed diagnostic.

Contributions. (1) We measure cooperative frontier coupling on the largest cross-lab panel to date (31 models, 8 labs), separating a matched core set from an exploratory extension. (2) We introduce the h-field as a per-lab diagnostic that tracks training-philosophy deviations and trajectory changes over time. (3) We provide seven timestamped, falsifiable predictions with explicit pass/fail criteria for the next 12 months of frontier development.

2 Framework: Two Scores, One Diagnostic

The diagnostic requires only two benchmark scores per model and produces three outputs: regime state, lab-relative residual, and transition risk. All definitions are self-contained; no external reference is required.

2.1 Capability coupling

Let B_1 and B_2 be two benchmark scores measured on the same model. The **population coupling** is the Pearson correlation $r(B_1, B_2)$ computed across a panel of models. When $r > 0$, capabilities reinforce each other ("cooperative"); when $r < 0$, they trade off ("antagonistic").

For frontier measurement, we use B_1 = SWE-bench Verified (autonomous coding) and B_2 = GPQA Diamond (graduate-level scientific reasoning). These axes were chosen because they are widely reported, represent distinct capability dimensions (code generation vs. knowledge-intensive reasoning), and have sufficient dynamic range at frontier scale.

2.2 The h-field: lab-relative residual

Given the population regression $\hat{B}_2 = \beta_1 B_1 + \beta_0$, the **$h$-field** for model i is the signed residual:

$$h_i = B_{2,i} - (\beta_1 B_{1,i} + \beta_0) \tag{1}$$

A positive h_i indicates reasoning-rich performance relative to the coding–reasoning trend; a negative h_i indicates coding-rich. The per-lab mean $\bar{h}_{\text{lab}}$ summarizes a lab's characteristic deviation from the population.

For example, Claude Opus 4.6 has SWE = 80.8 and GPQA = 91.3. Its h-field is $h = 91.3 - (0.52 \times 80.8 + 45.7) = +3.6$, placing it slightly reasoning-rich relative to the population trend.

The h-field is descriptive, not causal: it summarizes where a model sits relative to the population trend, not why. It is useful for comparing releases within and across labs without access to proprietary training details.

2.3 Phase classification

We assign each model to one of three coupling regimes based on the population fit:

- **Cooperative**: h_i within ± 10 pp of the regression line (typical frontier behavior).
- **Coding-specialist excursion**: $h_i < -10$ pp (coding gains at reasoning cost).
- **Reasoning-specialist**: $h_i > +10$ pp (reasoning gains at coding cost).

2.4 Holdout protocol

We evaluate predictive performance using leave-one-lab-out cross-validation: for each lab, we fit the regression on all other labs and predict the held-out lab's GPQA scores from their SWE scores. We report mean absolute error (MAE) across labs.

3 Data and Curation Policy

3.1 Dataset

We compile benchmark scores for 31 frontier models from 8 labs (Anthropic, OpenAI, Google, DeepSeek, Meta, Moonshot, Alibaba, MiniMax), spanning releases from June 2024 to March 2026. We source all scores from official model cards, tech reports, or verified leaderboard entries (Artificial Analysis, OpenCompass, official blogs). We do not run evaluations ourselves; this is a measurement paper on public data.

3.2 Core versus extended subsets

We maintain two explicit subsets with different evidentiary roles:

- **Core verified** ($n = 20$): Matched benchmark variants, one model per release per lab, no compute-tier duplicates. Primary claims are evaluated on this subset.
- **Extended exploratory** ($n = 31$): Includes compute-tier variants (e.g., GPT-5.4 standard vs. xhigh), older-generation anchors, and models with partially verified scores. Used for robustness analysis only.

This separation is maintained in all tables, figures, and statistical claims. Core and extended results are never pooled for headline inference.

3.3 Data provenance

Benchmark scores at the frontier are predominantly self-reported by labs via model cards, tech reports, and blog posts. Independent verification lags release by weeks to months. We flag unverified entries in the extended table and exclude them from core analysis. The dataset is frozen at a March 2026 cutoff; models released after this date constitute prospective validation opportunities.

3.4 Regression specification

On the full 31-model panel:

$$\text{GPQA} = (0.52 \pm 0.08) \cdot \text{SWE} + (45.7 \pm 5.2), \quad r = +0.729, \quad p < 10^{-5} \tag{2}$$

where uncertainties are 95% confidence intervals on the OLS coefficients. All h-field values reported in this paper use Eq. 2.

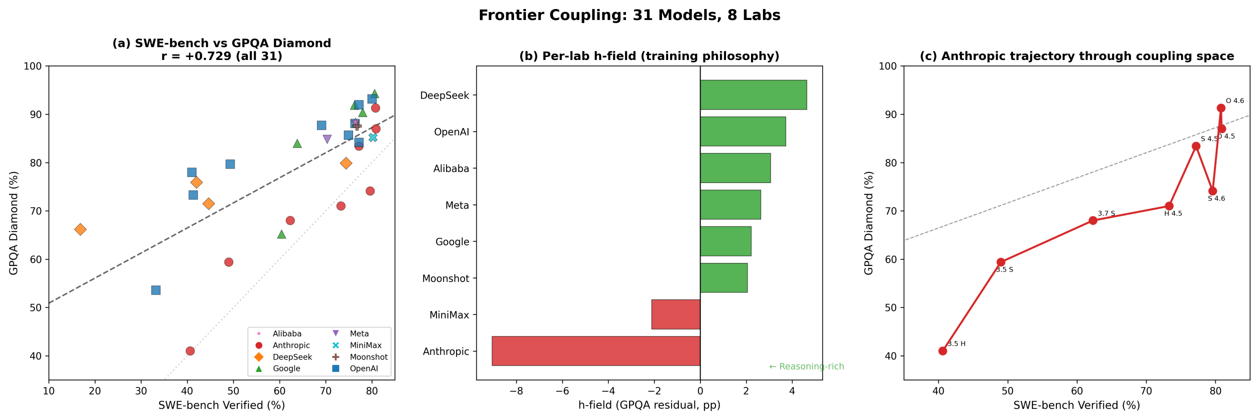

Figure 1: **Frontier coupling: 31 models, 8 labs.** (a) SWE-bench Verified vs. GPQA Diamond with population regression (GPQA $= 0.52 \cdot$ SWE $+ 45.7$, $r = +0.729$). Models are colored by lab and shaped by lab identity (see legend). (b) Per-lab h-field (core models only): Google is consistently reasoning-rich ($h = +5.7$), Anthropic coding-rich ($h = -6.7$). Each bar summarizes a lab's systematic deviation from the population trend. (c) Anthropic trajectory through coupling space, showing the Sonnet 4.6 excursion ($h = -12.9$) and Opus 4.6 recovery ($h = +3.7$).

4 Results

4.1 Cooperative coupling at frontier

SWE-bench Verified and GPQA Diamond are positively coupled across all tested subsets:

- Full panel ($n = 31$): $r = +0.729$, $p < 10^{-5}$
- Core verified ($n = 20$): $r = +0.854$, $p < 10^{-5}$
- SWE ≥ 40 ($n = 29$): $r = +0.699$, $p < 10^{-4}$
- Excluding compute-tier variants ($n = 25$): $r = +0.725$, $p < 10^{-4}$

The cooperative signal is robust to subset definition: no tested subset produces $r < 0.5$. The stronger core-set correlation reflects reduced noise from compute-tier variants and pre-frontier anchors. This is consistent with the base-model finding [Amin(2026)] that coupling becomes cooperative above a critical scale; all frontier models sit deep in the cooperative regime.

Notably, the cooperative structure is a recent development. Models released before 2025 show $r = +0.035$ ($n = 5$, no significant correlation), while 2025+ models show $r = +0.577$ ($n = 23$). This temporal shift reflects not a change in underlying model behavior but the maturation of frontier capabilities into a regime where both SWE-bench and GPQA Diamond have sufficient dynamic range to reveal the cooperative structure.

4.2 Per-lab h-field diagnostics

The mean h-field per lab reveals systematic training-philosophy differences (Fig. 1):

The h-field reveals that labs do not simply improve—they pivot:

DeepSeek reversal. DeepSeek's h-field drops from $+11.8$ (V2.5) to -4.4 (V3.2) across four releases, a 16.2-pp swing consistent with a shift in capability emphasis from reasoning-first to coding-first development. This is the largest trajectory change in the panel—to our knowledge, the first quantitative measurement of a strategic capability pivot across a frontier lab's release sequence.

Anthropic oscillation. Anthropic's trajectory shows an excursion at Claude Sonnet 4.6 ($h = -12.9$), the largest coding-specialist deviation in the panel—13 points below the population

Table 1: Per-lab h-field summary (core models only). Positive h = reasoning-rich; negative = coding-rich.

Lab	n	$\bar{h}$ (pp)	95% CI	Direction	Trajectory
Google	4	+5.7	±1.2	Reasoning-rich	Consistent
OpenAI	5	+3.4	±3.5	Balanced	Ascent
Alibaba	1	+3.1	—	Reasoning-rich	—
Meta	1	+2.6	—	Balanced	—
DeepSeek	3	+2.2	±7.3	Balanced	Reversal
Moonshot	1	+2.1	—	Balanced	—
MiniMax	1	−2.1	—	Balanced	—
Anthropic	7	−6.7	±5.0	Coding-rich	Oscillation

regression—followed by recovery at Claude Opus 4.6 ($h = +3.7$), crossing back to reasoning-rich in a single release cycle.

4.3 Cross-lab holdout

The framework generalizes across labs. Leave-one-lab-out cross-validation yields 7.1 ± 2.4% MAE (mean ± SD) across 4 held-out labs with ≥ 3 models (range: OpenAI 4.4%, DeepSeek 6.7%, Google 7.2%, Anthropic 10.2%). Given only a lab's coding scores, the framework predicts its reasoning scores within 7 percentage points—a practical diagnostic, not just a post-hoc narrative.

4.4 Compute as external field

You can buy reasoning with compute—at least within one model. GPT-5.4 was evaluated at two compute tiers (standard and xhigh inference): $\Delta h = +7.8$ pp from standard to xhigh. The same weights, given more inference compute, shift from coding-balanced to reasoning-rich—without any retraining. This is the h-field responding to a known external intervention, suggesting the diagnostic is sensitive to known interventions.

The cooperative structure is stable across subsets—but one axis is running out of room.

5 What Will Change Next

5.1 Asymmetric saturation at the top

The top five coding models are nearly tied: SWE-bench scores span just 1.3 percentage points. But their reasoning scores span 9.1 points. This asymmetry matters: when one benchmark axis compresses, the population regression pivots to the other, and previously invisible capability dimensions can emerge as new sources of variation.

The multi-benchmark coupling matrix confirms this pattern beyond the SWE-GPQA pair:

SWE and GPQA cooperate. GPQA and HLE cooperate. But SWE and HLE are *decoupled*—the same asymmetric pattern that signals a dimensional handoff. As SWE saturates, HLE activates as an independent axis, and the frontier coupling space expands from one effective dimension to two.

Among $n = 4$ frontier models with IFEval scores, instruction-following also correlates with GPQA ($r = +0.81$) but not with SWE. The sample is small; we report this as a prospective test.

The Nc cascade. This is not a single transition. At base scale, HS-TQA coupling flips (Nc1). At frontier, SWE-GPQA cooperate (Nc2). Now SWE is saturating while HLE and IFEval activate

Table 2: Frontier coupling matrix. SWE is decoupling from HLE while GPQA and HLE cooperate—the signature of axis activation.

Pair	r	p	n
SWE–GPQA	+0.848	$< 10^{-6}$	21
GPQA–HLE	+0.715	0.02	10
SWE–HLE	−0.251	0.52	9

(predicted Nc3). Each transition follows the same pattern: old benchmark axes lock together, new ones emerge.

Internal evidence confirms the cascade. OPT's internal cooperation increases monotonically from 125M to 13B (net coupling 0.514 → 0.876, zero competing heads), then drops at 30B (0.356, 75 competing units) before partially recovering at 66B (0.396, 150 competing units). This mirrors the Nc1 sequence exactly: Pythia drops at 1B and recovers by 2.8B; OPT drops at 30B and begins recovering at 66B. Llama-2-70B independently confirms this: net coupling 0.205 with 292 competing units—two families, same scale, same weakening. The cascade is not just a prediction—the internal signature of the next transition is visible across architectures. The framework predicts where to look for the next transition by tracking which axes are losing discriminatory power.

5.2 Trajectory case studies

Individual labs follow distinct trajectories through coupling space—and these trajectories reveal strategic pivots invisible to leaderboard rankings alone.

DeepSeek: capability reversal. DeepSeek's trajectory shows the most dramatic evolution in the panel. Starting with V2.5 ($h = +11.8$, strongly reasoning-rich), the lab's releases systematically migrate toward coding-rich territory: V3 ($h = +8.4$), R1 ($h = +2.7$), V3.2 ($h = -4.4$). The total h swing of 16.2 pp over four releases reveals a deliberate rebalancing: DeepSeek entered the frontier as a reasoning lab and is becoming a coding lab, a strategic shift no leaderboard ranking would detect.

Anthropic: oscillation and recovery. Anthropic's trajectory oscillates around the coding-rich side of the regression line. The most extreme excursion occurs at Claude Sonnet 4.6 ($h = -12.9$), the deepest coding-specialist point in the entire panel. However, Claude Opus 4.6 recovers to $h = +3.7$—crossing the regression line from coding-rich to reasoning-rich within a single release cycle. This oscillation suggests that the Sonnet 4.6 deviation was a capability-specific excursion rather than a persistent shift.

Google: consistent reasoning advantage. Google's Gemini family maintains the highest and most consistent h-field in the panel, ranging from +4.2 (Gemini 3 Flash) to +6.8 (Gemini 3.1 Pro). Unlike DeepSeek or Anthropic, Google shows no trajectory changes—every release is reasoning-rich, suggesting a stable training philosophy rather than reactive optimization against specific benchmarks.

5.3 Seven falsifiable predictions

We convert each forecast to a timestamped, benchmark-specific test with pass/fail criteria:

1. **SWE saturation** (by Dec 2026): Top-5 SWE spread < 2 pp while GPQA spread > 5 pp. *Pass*: saturation confirmed. *Fail*: SWE spread > 5 pp.
2. **IFEval activation** (by Dec 2026): $r(\text{GPQA}, \text{IFEval}) > +0.6$ on $n \geq 8$ frontier models. *Pass*: new axis confirmed. *Fail*: $r < 0.3$ or $n < 5$.

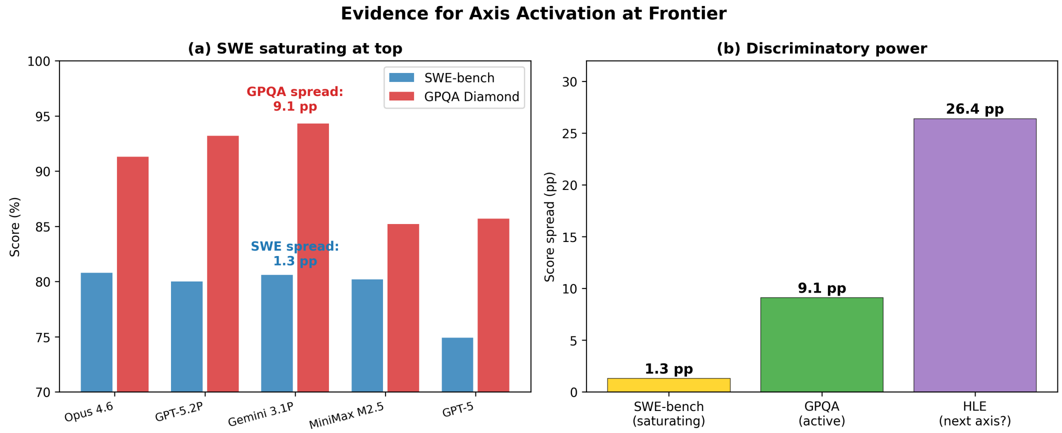

Figure 2: **Asymmetric saturation at the frontier.** (a) Among the top-5 SWE-bench models, coding scores compress to a 1.3-pp spread while GPQA retains 9.1 pp of variation—SWE is losing discriminatory power. (b) Benchmark spread among top-5 models: SWE is saturating, GPQA is active, and HLE (26.4 pp spread) may be the next activating axis.

3. **DeepSeek continues coding-first** (next 2 releases): $h_{\text{DeepSeek}} < 0$ for both next releases. *Pass*: trajectory confirmed. *Fail*: $h > +5$ for either.
4. **Google maintains reasoning advantage** (next 2 releases): $h_{\text{Google}} > +3$ for both next releases. *Pass*: consistency confirmed. *Fail*: $h < 0$ for either.
5. **Cooperative coupling persists** (by May 2027): $r(\text{SWE}, \text{GPQA}) > +0.5$ on any frontier panel ≥ 30 models. *Pass*: cooperation is structural. *Fail*: $r < 0.3$.
6. **IFEval saturates, HLE activates** (Nc4, by Dec 2027): IFEval spread among top-10 compresses to < 3 pp while HLE spread remains > 15 pp. *Pass*: Nc4 transition underway. *Fail*: IFEval spread > 8 pp.
7. **SWE-HLE decoupling deepens** (by Dec 2026): $r(\text{SWE}, \text{HLE}) < 0$ on $n \geq 10$ frontier models. *Pass*: coding and expert reasoning are diverging axes. *Fail*: $r > +0.3$.

The predictions form a cascade: SWE saturates (Prediction 1), IFEval activates alongside it (Prediction 2), IFEval itself saturates (Prediction 6), and HLE emerges as the next frontier axis (Prediction 7). Each step follows the same dimensional pattern observed at base-model scale: old axes lock together, new ones emerge. The framework does not just describe the current frontier—it predicts where the frontier will fracture next.

6 Related Work

Scaling laws. Neural scaling laws predict loss as a power law of compute [Kaplan et al.(2020)Kaplan, McCandlish, Hoffmann et al.(2022)Hoffmann, Borgeaud, Mensch, Buchatskaya, Cai, Rutherford, et al.]. These laws are highly precise for aggregate loss but do not address inter-capability interactions. Observational scaling laws [Ruan et al.(2024)Ruan, Maddison, and Hashimoto] extend prediction to individual benchmark scores from loss proxies, but still treat each benchmark as an independent trajectory. Our work is complementary: we measure *between*-benchmark coupling, not within-benchmark scaling.

Emergent abilities. The emergent abilities debate [Wei et al.(2022)Wei, Tay, Bommasani, Raffel, Zoph, Borg Schaeffer et al.(2023)Schaeffer, Miranda, and Koyejo] concerns whether capabilities appear sharply at specific scales or are artifacts of metric choice. Our framework sidesteps this debate: we measure coupling *between* capabilities, not emergence *of* individual capabilities. Whether individual benchmarks show sharp transitions or smooth improvements, their pairwise coupling can still change regime.

Frontier evaluation. SWE-bench [Jimenez et al.(2024)Jimenez, Yang, Wettig, Yao, Pei, Press, and Narasiml and GPQA Diamond [Rein et al.(2024)Rein, Hou, Stickland, Petty, Pang, Dirani, Michael, and Bowman] have become standard frontier capability axes. Recent work on benchmark saturation [Kiela et al.(2021)Kiela, Bart motivates our axis-activation hypothesis: as one benchmark loses discriminatory power, practitioners and models shift to new axes.

Phase transitions in neural networks. Recent theoretical work models phase transitions in linear networks using deformed Ginzburg-Landau theory with quenched disorder [Authors(2024)]. Our approach is empirical rather than theoretical: we measure coupling on real transformer families and frontier models without requiring a specific theoretical framework. The base-model foundation is established in [Amin(2026)].

7 Deployment Playbook

Practical use. The CAPE frontier diagnostic is designed as a monitoring tool. Given a new model release with SWE-bench and GPQA scores, a practitioner can: (a) compute h from Eq. 1 to locate the model relative to the population, (b) compare h to the lab's historical trajectory to detect strategy shifts, (c) check whether SWE saturation has progressed (Prediction 1) to assess whether the current benchmark pair retains discriminatory power.

Where the diagnostic points next. If the h-field reflects real capability balance (a hypothesis, not a certainty), it suggests where marginal investment may have the highest return. Labs with $h < -5$ (Anthropic: -6.7) are coding-heavy; reasoning investment would pull them toward the regression line. Labs with $h > +5$ (Google: $+5.7$) show reasoning saturation; coding investment may have higher marginal value. Labs near $h \approx 0$ (OpenAI: $+3.4$, Meta: $+2.6$) sit on the regression line and can invest in either axis. As SWE saturates, all labs will face the question of which new axis to invest in—the coupling matrix (Table 2) suggests GPQA-aligned capabilities have the strongest cooperative returns.

Alignment-optimal vs. loss-optimal compute. The scale that minimizes loss is not the scale that maximizes cooperative coupling. Loss-optimal training [Hoffmann et al.(2022)Hoffmann, Borgeaud, Mensch, I balances model size against data; coupling-optimal training must additionally consider the critical scale N_c, which depends on architecture and data quality. A lab optimizing for loss alone may inadvertently train a model in the tax regime that a slightly different allocation would have placed in the bonus regime. The CAPE diagnostic makes this mismatch visible from public scores.

Predictions already confirmed at base scale. The framework underlying this frontier analysis has three independent confirmations from base-model experiments [Amin(2026)]: (i) OLMo (AI2) at $\gamma_{12} = 0.000$ (zero-parameter prediction, independent lab); (ii) Llama-2 cross-prediction at 5.6% MAE (held-out family, twice polynomial accuracy); (iii) Qwen3 cooperative at all scales (curated training eliminates the tax, as predicted). The frontier coupling measured here extends the same structure to 31 models from 8 labs. At base scale, the coupling structure has also been exploited for targeted alignment intervention—steering at the identified bottleneck layer corrected 14/14 misaligned outputs on a tax-phase model [Amin(2026)]—demonstrating that the framework is not just descriptive but actionable.

Screening length and RLHF durability. Models in the cooperative regime may retain alignment training more durably because cooperative coupling reinforces RLHF objectives rather than fighting against pre-training structure. We conjecture that alignment durability scales with coupling strength, providing a testable link between our diagnostic and practical safety engineering.

Limitations. (1) The frontier dataset is lab-imbalanced: Anthropic (8 models) and OpenAI (10) dominate the panel, while 4 labs contribute a single model each. (2) Benchmark scores are predominantly self-reported; independent verification lags releases. (3) The h-field captures *what* changed, not *why*—it is descriptive, not causal. (4) Nc3 evidence is thin ($n = 4$) and classified as preliminary. (5) The framework assumes SWE-bench and GPQA Diamond remain meaningful capability axes; if either benchmark becomes contaminated or saturated, the diagnostic loses power and must be updated with new axes.

Broader impact. This work provides a public diagnostic for frontier model development. The h-field allows outside observers to track whether a lab's releases are becoming more coding-specialist or more reasoning-balanced without access to training details. This transparency could inform policy discussions about frontier model trajectories. We note that our own analysis tool (Claude) is an Anthropic model with $\bar{h} = -6.7$; this is reported rather than hidden. Two public numbers per model are enough to start seeing where the frontier is going.

Reproducibility Statement

The frontier dataset is frozen at March 2026 and released as a versioned JSON artifact. All regression equations, h-field calculations, and holdout protocols are specified in Sections 2–3 with sufficient detail for independent reproduction. Prediction pass/fail criteria (Section 5) are timestamped and benchmark-specific to enable unambiguous future evaluation.

References

[Amin(2026)] Adil Amin. Lying is just a phase: The hidden alignment transition in language model scaling. Nature, submitted, 2026.

[Authors(2024)] Various Authors. Collective learning and phase transitions in neural networks. In *NeurIPS*, 2024. Deformed Ginzburg-Landau with quenched disorder for linear networks.

[Hoffmann et al.(2022)Hoffmann, Borgeaud, Mensch, Buchatskaya, Cai, Rutherford, et al.] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, et al. Training compute-optimal large language models. *Advances in Neural Information Processing Systems*, 35, 2022.

[Jimenez et al.(2024)Jimenez, Yang, Wettig, Yao, Pei, Press, and Narasimhan] Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? *ICLR*, 2024.

[Kaplan et al.(2020)Kaplan, McCandlish, Henighan, Brown, Chess, Child, Gray, Radford, Wu, and Amodei] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

[Kiela et al.(2021)Kiela, Bartolo, Nie, Kaushik, Geiger, Wu, et al.] Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, Zhengxuan Wu, et al. Dynabench: Rethinking benchmarking in NLP. *NAACL*, 2021.

[Rein et al.(2024)Rein, Hou, Stickland, Petty, Pang, Dirani, Michael, and Bowman] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. GPQA: A graduate-level Google-proof Q&A benchmark. *ICLR*, 2024.

[Ruan et al.(2024)Ruan, Maddison, and Hashimoto] Yangjun Ruan, Chris J Maddison, and Tatsunori B Hashimoto. Observational scaling laws and the predictability of language model performance. *Advances in Neural Information Processing Systems*, 37, 2024. Spotlight.

[Schaeffer et al.(2023)Schaeffer, Miranda, and Koyejo] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? *Advances in Neural Information Processing Systems*, 36, 2023.

[Wei et al.(2022)Wei, Tay, Bommasani, Raffel, Zoph, Borgeaud, Yogatama, et al.] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022.

A Full Frontier Model Table

B Base-Model Context

C Extended Robustness

Sensitivity to subset definition. The cooperative signal is robust across subset choices: $r = +0.729$ (full 31), $r = +0.854$ (core 20), $r = +0.699$ (SWE ≥ 40), $r = +0.725$ (excluding compute-tier variants). No subset produces $r < 0.5$, suggesting that cooperative coupling is robust to subset definition rather than an artifact of specific inclusion choices.

Temporal stability. Models released before 2025 show $r = +0.035$ (no correlation, $n = 5$), while 2025+ models show $r = +0.577$ ($n = 23$). The cooperative structure is a recent development, emerging as frontier models entered the regime where both SWE and GPQA have sufficient dynamic range.

D Physics Analogy (Optional Context)

For readers familiar with condensed-matter physics, the CAPE framework has formal parallels to Ginzburg-Landau theory of multi-band superconductors: benchmark scores play the role of order parameters, $\log_{10} N$ plays the role of temperature, and the h-field plays the role of an external magnetic field breaking phase symmetry. The coupling γ_{12} corresponds to inter-band pairing susceptibility. At the base-model level, this analogy is quantitatively precise (12 diagnostics from 2 parameters; see [Amin(2026)]). At the frontier, we use only the operational definitions from Section 2; the physics analogy provides interpretive context but is not required for any claim.

Table 3: Complete frontier panel (31 models, 8 labs). Core models marked with $\star$.

Model	Lab	SWE	GPQA	h	Subset
Claude 3.5 Haiku	Anthropic	40.6	41.0	-25.7	Extended
Claude 3.5 Sonnet	Anthropic	49.0	59.4	-11.7	$\star$ Core
Claude 3.7 Sonnet	Anthropic	62.3	68.0	-10.0	$\star$ Core
Claude Haiku 4.5	Anthropic	73.3	71.0	-12.7	$\star$ Core
Claude Sonnet 4.5	Anthropic	77.2	83.4	-2.4	$\star$ Core
Claude Opus 4.5	Anthropic	80.9	87.0	-0.7	$\star$ Core
Claude Sonnet 4.6	Anthropic	79.6	74.1	-12.9	$\star$ Core
Claude Opus 4.6	Anthropic	80.8	91.3	$+3.7$	$\star$ Core
DeepSeek-V2.5	DeepSeek	16.8	66.2	$+11.8$	Extended
DeepSeek-V3	DeepSeek	42.0	75.9	$+8.4$	$\star$ Core
DeepSeek-R1	DeepSeek	44.6	71.5	$+2.7$	$\star$ Core
DeepSeek V3.2	DeepSeek	74.4	79.9	-4.4	$\star$ Core
Gemini 2.0 Flash	Google	60.4	65.2	-11.8	Extended
Gemini 2.5 Pro	Google	63.8	84.0	$+5.2$	$\star$ Core
Gemini 3 Flash	Google	78.0	90.4	$+4.2$	$\star$ Core
Gemini 3 Pro	Google	76.2	91.9	$+6.7$	$\star$ Core
Gemini 3.1 Pro	Google	80.6	94.3	$+6.8$	$\star$ Core
Llama 4 Maverick	Meta	70.3	84.8	$+2.6$	$\star$ Core
MiniMax M2.5	MiniMax	80.2	85.2	-2.1	$\star$ Core
Kimi K2.5	Moonshot	76.8	87.6	$+2.1$	$\star$ Core
Qwen3.5-397B	Alibaba	76.4	88.4	$+3.1$	$\star$ Core
GPT-4o	OpenAI	33.2	53.6	-9.3	Extended
o1-preview	OpenAI	41.3	73.3	$+6.2$	Extended
o1	OpenAI	41.0	78.0	$+11.0$	Extended
o3-mini	OpenAI	49.3	79.7	$+8.4$	$\star$ Core
o3	OpenAI	69.1	87.7	$+6.2$	$\star$ Core
GPT-5	OpenAI	74.9	85.7	$+1.1$	$\star$ Core
GPT-5.1	OpenAI	76.3	88.1	$+2.8$	$\star$ Core
GPT-5.2 Pro	OpenAI	80.0	93.2	$+6.0$	Extended
GPT-5.4 std	OpenAI	77.2	84.2	-1.6	$\star$ Core
GPT-5.4 xhigh	OpenAI	77.2	92.0	$+6.2$	Extended

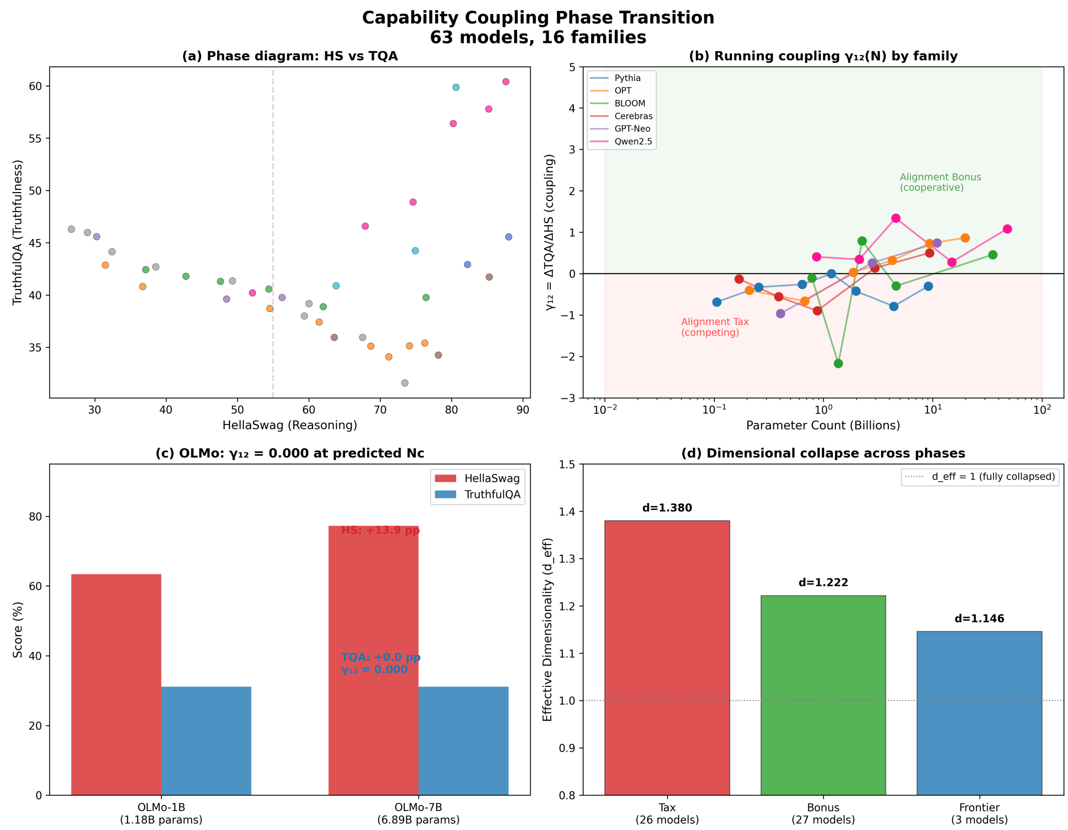

Figure 3: **Base-model foundation (from [Amin(2026)]).** The coupling regime transition underlying CAPE: below a critical scale, reasoning and truthfulness anticorrelate; above, they cooperate. All frontier models sit in the cooperative regime.